

PROJECT REPORT

Netflix Movie Recommendation System Using Content-Based Filtering

1. Introduction

With the rapid growth of online streaming platforms such as Netflix, users are exposed to thousands of movies and TV shows. Finding relevant content from such a large catalog can be challenging. Recommendation systems play a vital role in improving user experience by suggesting content based on user preferences.

This project focuses on building a **content-based movie recommendation system** that recommends movies similar to a given movie by analyzing metadata such as genre, cast, director, and description. The system uses Natural Language Processing (NLP) and similarity-based techniques to generate personalized recommendations.

2. Dataset Description

The dataset used in this project is the **Netflix Movies and TV Shows Dataset** obtained from Kaggle. It contains information about movies and TV shows available on Netflix.

Important Columns

- show_id
- title
- director
- cast
- country
- release_year
- rating
- duration
- listed_in (Genre)
- description

The dataset includes both movies and TV shows, with some missing values in columns such as director and cast, which were handled during preprocessing.

3. Methodology

3.1 Data Preprocessing

- Missing values were handled by replacing them with empty strings.
- Relevant text features such as **title, genre, cast, director, and description** were selected.
- Text cleaning included:
 - Lowercasing
 - Removing punctuation
 - Removing stopwords

3.2 Feature Engineering

- A **metadata soup** was created by combining selected text features.
- Text data was converted into numerical vectors using:
 - **TF-IDF Vectorizer**
 - (Optional) Count Vectorizer

3.3 Similarity Computation

- **Cosine similarity** was calculated between all movie vectors.
 - Movies with the highest similarity scores were selected as recommendations.
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4. Recommendation System

A function was created that:

- Takes a movie title as input
- Finds similar movies based on cosine similarity
- Returns **Top-N recommended movies**

This allows users to receive personalized recommendations based on their interests.

5. Results and Evaluation

- The recommendation system successfully suggests movies with similar genres and themes.
 - Manual validation confirmed that recommendations are relevant and meaningful.
 - Different feature combinations were tested to improve recommendation quality.
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6. Conclusion

This project demonstrates how content-based filtering can be used to build an effective movie recommendation system. By leveraging NLP techniques and cosine similarity, the system provides personalized movie suggestions without requiring user ratings.

Such recommendation systems improve user engagement and content discovery and form the foundation of modern streaming platforms like Netflix.